



National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

CS 1005 : Discrete Structures

Assignment 2

Instructor : Ms Mahzaib Younas

SECTION BCS (C)

Q No 01: 200 students can use bus service to go to the university in buses, X, Y, and Z

50 students did not use the bus service.
8 students came to university on bus X.
22 students came to university on bus Y.
19 students came to university on bus Z.
10 students used all buses X, Y and Z
25 students used bus X and Z.

1. Draw the Venn diagram with all sets enumerated as far as possible, Label the subset which cannot be enumerated as x and y in any order.
2. If half as many students in bus X as Z write down the pair of simultaneous equations in x and y.
3. Solve the equation and find the value of x and y.

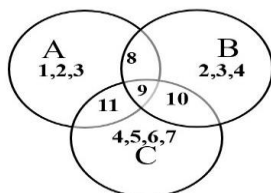
QNO02: Proving the set identities by using membership table.

1. $(A \cup B) - C = (A - C) \cup (B - C)$
2. $(A \cap B)^c = A^c \cup B^c$

QNO03: prove the following by applying the laws, and also draw the Venn diagram
 $(A - B) - C = (A - C) - B.$

QNO04: Use the Venn diagram given below and find the following.

1. $(A \cap B)^c - C$
2. $A^c \cup (B \cup C)$



QNO05: If A and B are the subset of universal set then prove the set difference law.

QNO06: Which of the following are true statements?

1. $2 \in \{1, 2, 3\}$	4. $\{2\} \subseteq \{1, 2, 3\}$
2. $\{2\} \in \{1, 2, 3\}$	5. $\{2\} \subseteq \{\{1\}, \{2\}\}$
3. $2 \subseteq \{1, 2, 3\}$	6. $\{2\} \in \{\{1\}, \{2\}\}$



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QNO06: 100 students were asked which fast food restaurant they have been to this year. The results of the survey were as follows: 5 have been to all three 20 have been to both McDonald's and Taco Bell 25 have been to both Subway and McDonald's 15 have been to both Taco Bell and Subway 50 have been to McDonald's 40 have been to Taco Bell 45 have been to Subway.

- a) Create a Venn diagram to model the information.
- b) How many did not go to any of the three?
- c) How many have been to McDonald's or Taco Bell?
- d) How many have been to McDonald's or Taco Bell, but not Subway?
- e) How many have been to exactly two of the three?

QNO07: Suppose that the universal set is $U = \{1,2,3,4,5,6,7,8,9,10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise. ($2 * 2 = 4$)

- a) $\{2,3,4,7,8,9\}$
- b) $\{3,4,5\}$